

27.12.23

MathsAns 1.

$$210 = 5 \times 2 \times 3 \times 7$$

$$55 = 5 \times 11$$

$$\text{HCF} = 5$$

$$210 \times 5 + 55y = 5$$

$$1050 + 55y = 5$$

$$55y = 5 - 1050$$

$$y = \frac{-1045}{55}$$

$$y = -19$$

(1) -19

Ans 2.

$$a+b = 3+7 = 10$$

$$10 = 2 \times 5$$

least prime factor is 2.

(1) 2

Ans 3.

Smallest two digit composite no. = 10

Smallest composite no. = 4

$$4 = 2 \times 2$$

$$10 = 2 \times 5$$

$$\text{LCM} = 2 \times 2 \times 5$$

$$= 20$$

(3) 20

Ans 4.

$$18 - 7 = 11$$

$$21 - 10 = 11$$

$$24 - 13 = 11$$

$$18 = 2 \times 3 \times 3$$

$$21 = 3 \times 7$$

$$24 = 2 \times 2 \times 2 \times 3$$

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 7$$

$$= 8 \times 9 \times 7$$

$$= 504$$

$$504 - 11 = 493 //$$

$$(2) 493$$

Ans 5.

$$p(y) = y^5 - 3y^4 + 2y^2 + 6$$

$$y = 1$$

$$= 1^5 - 3(1)^4 + 2(1)^2 + 6$$

$$= 1 - 3 + 2 + 6$$

$$= -2 + 2 + 6$$

$$= 6$$

$$q(y) = y^5 - y^3 + 2y^2 + y - 6$$

$$y = 2$$

$$= (2)^5 - (2)^3 + 2(2)^2 + 2 - 6$$

$$= 32 - 8 + 8 + 2 - 6$$

$$= 42 - 14$$

$$= 28$$

$$p(1) = 6, q(2) = 28$$

$$p(1) + q(2) = 6 + 28 = 34 //$$

$$(1) 34$$

Ans 6.

Let the zeroes be α and β

$$\alpha = \beta$$

The zeroes are equal so,

$$b^2 - 4ac = 0$$

$$b = 3x,$$

$$a = 2$$

$$c = 8$$

$$\text{So, } (3x)^2 - 4(2)(8) = 0$$

$$9x^2 = 64$$

$$x^2 = \frac{64}{9}$$

$$x = \sqrt{\frac{64}{9}}$$

$$x = \pm \frac{8}{3} //$$

$$(4) \pm \frac{8}{3}$$

Ans 7.

Ans 8.

Ans 9.

Ans 7: we know, $ax^3 + bx^2 + cx + d = 0$

So, $(1) \quad by = \frac{c}{1}$

$by = -c \Rightarrow -by = c \quad \text{--- (1)}$

and

$ax^3 + bx^2 + cx + d = 0$

$(-1)(b) + (-c) + (-1)(d) = 0$

$-b - c - d = 0$

$b + c + d = 0$

Equating (1) and (2),

$-by = b - a + 1$

$by = -b + a - 1$

$by = a - b - 1$

(4) $a - b - 1$

Ans 8: $x^2 - 5x + 1 = 0$

dividing both sides by x ,

$x - 5 + \frac{1}{x} = 0$

$x + \frac{1}{x} = 5$

(1) 5

Ans 9: given, $b^2 - 4ac = -68$

$a = m$

$b = -4$

$c = 7$

$-68 = (-4)^2 - 4(m)(7)$

$-68 = 16 - 28m$

$-68 - 16 = -28m$

$-84 = -28m$
 $m = 3$

(4) 3

Ans 10. (3) $x^2 - 6x - 5 = 0$

Ans 11. let the integers be x and $x+1$.

ATQ, $(x+x+1)^2 = 365$

(1) $(2x+1)^2 = 365$

$4x^2 + 4x + 1 = 365 = 0$

$4x^2 + 4x - 364 = 0$

By 4,

$x^2 + x - 91 = 0$

$x^2 + (x+1)^2 = 365$

$x^2 + x^2 + 2x + 1 = 365$

$2x^2 + 2x + 1 - 365 = 0$

$2x^2 + 2x - 364 = 0$

By 2,

$x^2 + x - 182 = 0$

$x^2 + 14x - 13x - 182 = 0$

$x(x+14) - 13(x+14) = 0$

$(x-13)(x+14) = 0$

So, $x = 13$ and $x = -14$

It's a positive integer

So, the no. is 13 and the consecutive no. is 14. //

Ans 12. $a + \beta = \frac{-b}{a}$ and $a\beta = \frac{c}{a}$

$x^2 - px - p - k = 0$

So,

$a + \beta = \frac{-(-p)}{1} = p$

$a\beta = \frac{-p-k}{1} = -p-k$

$(a+1)(\beta+1) = 6$

$a\beta + a + \beta + 1 = 6$

Ans 13. let

Ans 14.

$$a \beta + a + \beta = 5$$

$$p - p - k = 5$$

$$-k = 5$$

$$k = -5$$

Ans 13. let his present age be x ,

$$\frac{1}{x-3} + \frac{1}{x+5} = \frac{1}{3}$$

$$\frac{x+5+x-3}{(x-3)(x+5)} = \frac{1}{3}$$

$$\frac{2x+2}{x^2-3x+5x-15} = \frac{1}{3}$$

$$3(2x+2) = x^2-2x-15$$

$$6x+6-x^2-2x-15=0$$

$$-x^2+4x-9=0$$

$$x^2-4x-21=0$$

$$x^2-7x+3x-21=0$$

$$x(x-7)+3(x-7)=0$$

$$(x+3)(x-7)=0$$

$$\text{So, } x = -3 \text{ and } x = 7$$

Age can't be negative so, $x = 7$

~~∴ This present age is 7.~~

Ans 14. $40 = 2 \times 2 \times 2 \times 5$

$$42 = 2 \times 3 \times 7$$

$$45 = 3 \times 3 \times 5$$

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7$$

$$= 8 \times 9 \times 35$$

$$= 72 \times 35$$

$$= 2520 \text{ cm}$$

Therefore each should walk 2520 cm so that each can cover the same distance in complete steps. //

Ans 15. (i) By contradiction,

let us assume $\sqrt{5}$ is rational

So, $\sqrt{5} = \frac{a}{b}$ [where a and b are ~~not~~ co-primes and integers and $b \neq 0$.]

$$\sqrt{5} b = a$$

Squaring both sides,

$$5b^2 = a^2 \quad \text{--- (1)}$$

$$b^2 = \frac{a^2}{5}$$

By theorem:- If p divides a^2 it also divides a .

$$\Rightarrow \frac{a}{5}, \quad 5 \text{ is a factor of } a \quad \text{--- (2)}$$

Now let $\frac{a}{5} = c$ (c is any positive integer).

$$a = 5c$$

Putting in (1),

$$5b^2 = (5c)^2$$

$$5b^2 = 25c^2$$

$$b^2 = 5c^2$$

$$255$$

$$\frac{b^2}{5} = c^2$$

By using the same above theorem,

$$\Rightarrow \frac{b}{5}, \quad \text{so, } 5 \text{ is a factor of } b \quad \text{--- (3)}$$

From (2) and (3)

So, a and b are ~~not~~ not co prime nos and hence our assumption is wrong and

$\sqrt{5}$ is an irrational no.

Hence proved!!

(ii) let a be the positive integer and $b = 3$
 we know, $a = bq + r$,
 $0 \leq r < b$

Now, $a = 3q + r$, $0 \leq r < 3$

The possibilities of remainder is 0, 1 or 2.

Case 1:- when $a = 3q$

$$a^2 = (3q)^2 = 9q^2 = 3q \times 3q = 3m$$

where $m = 3q^2$

Case 2:- when $a = 3q + 1$

$$a^2 = (3q + 1)^2$$

$$= 9q^2 + 6q + 1$$

$$= 3q(3q + 2) + 1$$

$$= 3m + 1$$

where $m = q(3q + 2)$

Case 3:- when $a = 3q + 2$

$$a^2 = (3q + 2)^2$$

$$= 9q^2 + 12q + 4$$

$$= 3q(3q + 4) + 1$$

$$= 3m + 1$$

Hence proved!

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$$2x^2 + x - 2 \sqrt{\quad}$$

$$4x^4 - 2x^3 - 6x^2 + x - 5$$

$$4x^4 + 2x^3 + 2x^2$$

$$\ominus \quad (+) \quad (+)$$

$$0 - 4x^3 - 8x^2 + x$$

$$- 2x^3 - 2x^2 + 4x$$

$$(+ \quad) \quad (+) \quad (-)$$

$$- 2x^3 - 6x^2 - 3x$$

$$- 2x^3 - x^2 + 2x$$

$$(+ \quad) \quad (+) \quad (-)$$

$$0 - 5x^2 - 5x - 5$$

$$\oplus \quad -4x^2 - 2x + 4$$

$$x^2 - 3x - 9$$

$$Q = 2x^2 - 3x - 2$$

$$R = x^2 - 3x - 9$$

$\therefore x^2 - 3x - 9$ should be added. //

(ii) zeroes = $2 \pm \sqrt{3}$
 $(x - 2 + \sqrt{3})$ and $(x - 2 - \sqrt{3})$

$$= (x - 2 + \sqrt{3})(x - 2 - \sqrt{3})$$

$$= x^2 + 4 - 4x - 3$$

$$= x^2 - 4x + 1$$

$$\begin{array}{r} x^2 - 2x - 35 \\ x^2 - 4x + 1 \end{array} \quad \begin{array}{r} x^4 - 6x^3 - 26x^2 + 138x - 35 \\ x^4 - 4x^3 - x^2 \end{array}$$

$$\begin{array}{r} -2x^3 - 27x^2 + 138x - 35 \\ -2x^3 + 8x^2 - 2x \end{array}$$

$$\begin{array}{r} -35x^2 + 140x - 35 \\ -33x^2 + 140x - 35 \end{array}$$

$$\begin{array}{r} -33x^2 + 140x - 35 \\ 0 \end{array}$$

So,

$$x^2 - 2x - 35 = 0$$

$$x^2 - 7x + 5x - 35 = 0$$

$$x(x - 7) + 5(x - 7) = 0$$

$$(x + 5)(x - 7) = 0$$

So,

$$x = -5 \text{ and } x = 7$$

Hence other zeroes are -5 and 7 . //

Ans 18. (i) (1.) -2, 8

(ii) (2.) $-x^2 + 6x + 16$

(iii) $= -x^2 + 6x + 16$

$x = 4$

$= -(4)^2 + 6(4) + 16$

$= -16 + 24 + 16$

$= 24$

(3.) 24

(iv) $= -x^2 + 3x - 2$

$= x^2 - 3x + 2$

$= x^2 - 2x - x + 2$

$= x(x-2) - 1(x-2)$

$= (x-1)(x-2)$

So, $x = 1$ and 2

(4.) 1, 2

(v) $a + b = 4 - 3 = 1$

$ab = -12$

So, $x^2 - x + 24$

(1.) $x^2 - x + 24$

————— X —————